



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PONNI COOPERATIVE PETROLEUM.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

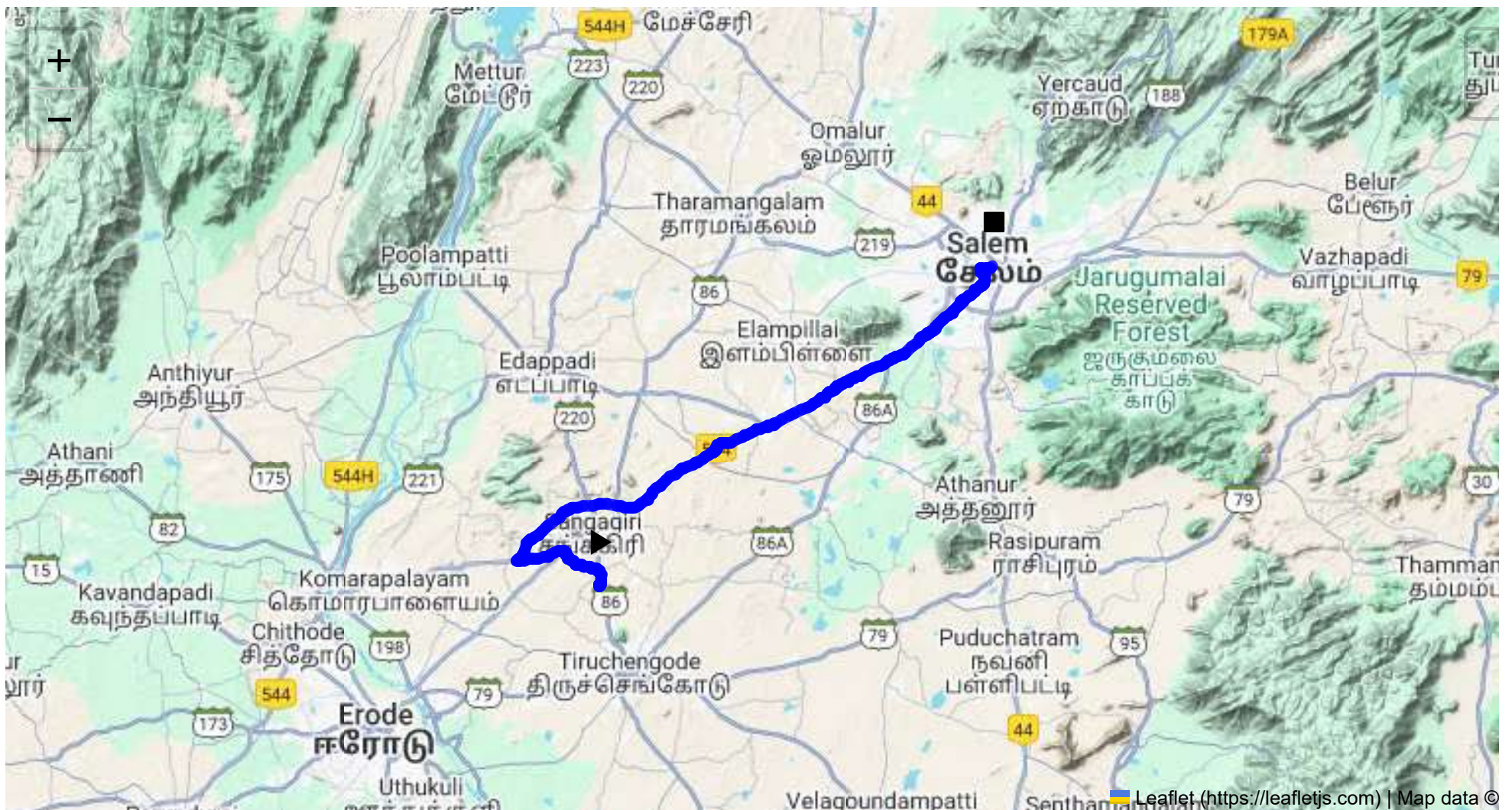
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 53.75 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.657359, 78.148842)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route from CVQF+23W, Sangagiri to Lock No IG7103 in Salem via Sankari and other intermediate locations covers approximately 53.75 kilometers. Key segments include Sankari RS - Erode Privu Rd, Bhavani Main Road, and NH544. The final approach through Salem involves urban roads with varying degrees of congestion and infrastructure.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

Tamil Nadu generally experiences tropical weather. The region has hot and dry conditions from March to June, the southwest monsoon from July to September, and the northeast monsoon from October to December. Potential hazards include heavy rainfall and flooding during the monsoon months, which could affect road conditions and visibility, increasing the risk of hydroplaning.

3. Analysis of Traffic Patterns:

Traffic is typically most congested during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) peak hours, especially near urban areas like Salem. Bhavani Main Road and sections near Sangagiri are prone to congestion. NH544 is a major highway and can experience high traffic volumes, particularly on weekends and holidays.

4. Assessment of Road Quality and Infrastructure:

The NH544 is generally well-maintained with decent road quality but may have occasional construction work and repairs. Rural roads such as those in Sankari may have uneven surfaces and narrower lanes. Urban stretch roads in Salem might encounter more potholes and uneven patches which can be challenging for heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies:

In case of emergencies, alternate routes include taking the Sankagiri - Tiruchengode Road if entering from the western region, or Erode for a detour. Local routes like Edappadi Road offer detours in rural segments but require careful navigation.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

Transport of hazardous materials in Tamil Nadu is regulated under the Motor Vehicles Act, with requirements for special permits, indicated timings (usually not allowed during peak hours), and specific routes for transport. Regular checks by road transport authorities are expected.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

While specific local incidents may not be widely documented, heavy vehicular incidents generally involve speeding, mechanical failures, or loss of vehicle control, particularly on bends and slopes around hilly areas approaching Salem.

8. Environmental Considerations and Sensitive Areas:

This region has pockets of agricultural lands and proximity to reserve forests, particularly near Salem. Efforts should be taken to avoid spills and emissions. Nearby water bodies should be safeguarded from contamination due to hazardous materials.

9. Analysis of Communication Coverage:

The route has decent mobile network coverage except in rural pockets where signal strength may be weak, particularly around Sankari and certain areas approaching Salem. Necessary pre-trip communication checks are advisable.

10. Estimated Emergency Response Times for Different Route Segments:

Response times can vary. In urban Salem, response might be quicker (15-30 minutes) due to better infrastructure and emergency services. Rural areas might face longer delays (up to an hour) due to sparse services and challenging access.

12. Overall Summary of Risk Assessment:

The route presents manageable risks but requires vigilance, particularly with weather disruptions and urban traffic congestion. Critical segments near Sankari and Salem need heightened attention. Preparedness in terms of vehicle condition, permit compliance, and emergency readiness is essential to mitigate risks. Regular updates on weather forecasts and traffic advisories can assist in optimizing transit safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.46048, 77.85036	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45634, 77.81673	15 KM/Hr
11	Turn	High	11.61233, 78.10397	15 KM/Hr
12	Turn	Medium	11.61216, 78.10423	30 KM/Hr
13	Turn	Medium	11.65141, 78.14234	30 KM/Hr
14	Turn	Medium	11.65210, 78.14169	30 KM/Hr
15	Blind Spot	Blind Spot	11.65573, 78.13972	10 KM/Hr
16	Turn	Medium	11.65636, 78.14508	30 KM/Hr
17	Turn	High	11.65678, 78.14605	15 KM/Hr
18	Turn	Medium	11.65674, 78.14623	30 KM/Hr
19	Turn	High	11.65628, 78.14659	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.65626, 78.14667	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	clinic	Salem Care Foundation	11.6371946, 78.1313713	30 km/h	Medium
19	hospital	Aravind Eye Hospital	11.641927, 78.1392347	30 km/h	Medium
20	police	Annadanapatti B 4 Police Station	11.6424031, 78.1404779	30 km/h	Medium
21	clinic	Shri Bharthi Nursing Home	11.6420069, 78.1424099	30 km/h	Medium
22	hospital	Pramilagurunathan Hospital	11.6544063, 78.1464484	30 km/h	Medium
23	hospital	Nalam Hospital	11.65862, 78.147448	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	hospital	Shree Varma Ayurveda Hospital	11.656905, 78.14717	30 km/h	Medium
25	hospital	London Ortho Speciality Hospital	11.6597962, 78.1507714	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673

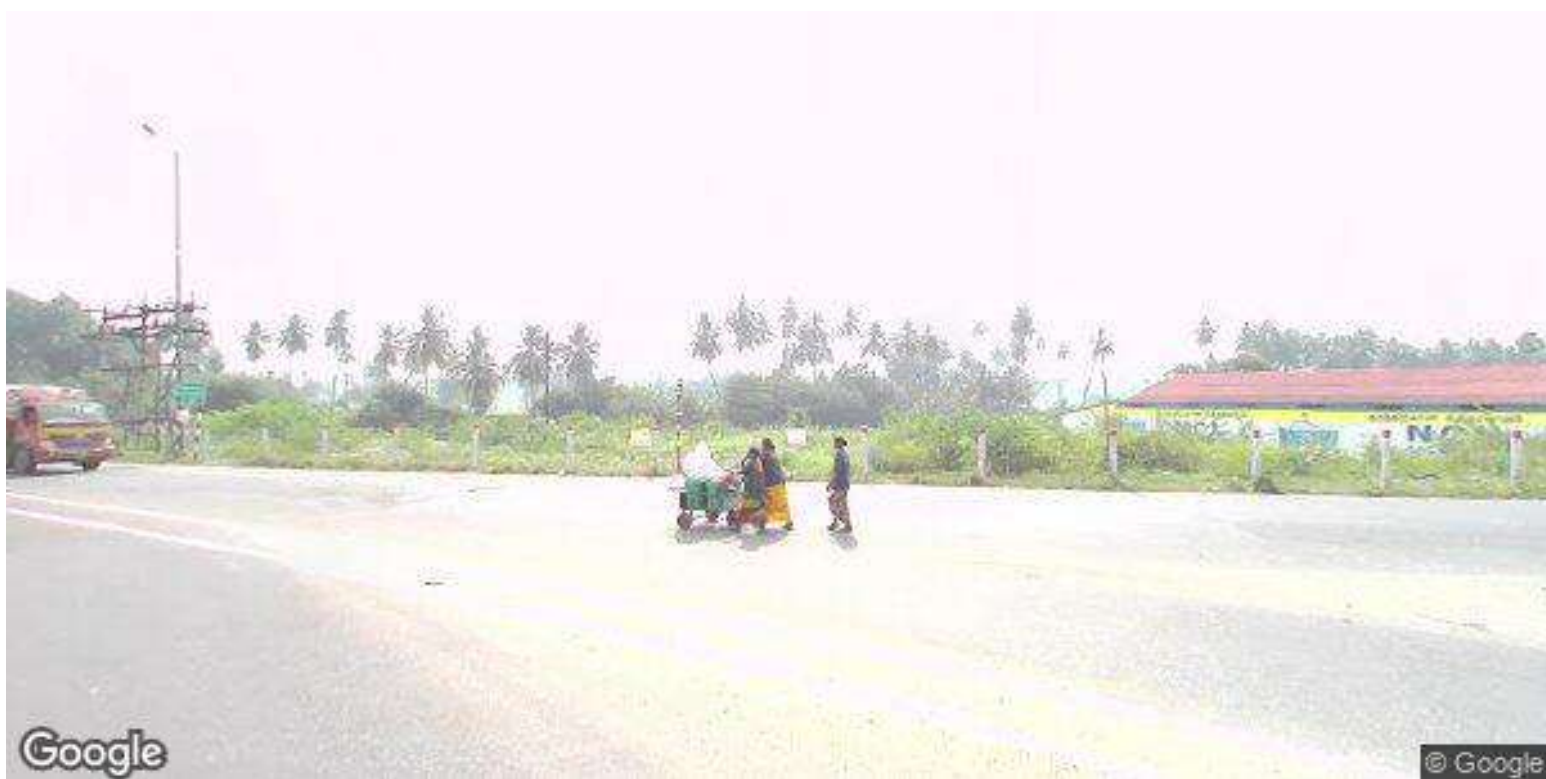


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61216, 78.10423



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65141, 78.14234



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65210, 78.14169



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.65573, 78.13972



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65636, 78.14508



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65678, 78.14605



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65674, 78.14623



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65628, 78.14659



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65626, 78.14667
